

6 The sinoatrial node (SAN), the atrioventricular node (AVN) and the Purkyne fibres have a role in the initiation and control of heart action.

(a) The sinoatrial nodal artery (SN artery) is a branch of one of the main arteries serving the cardiac muscle. Partial blockage of the SN artery as a result of cardiovascular disease can cause the SAN to malfunction.

(i) Name the main artery serving the cardiac muscle from which the SN artery branches.

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(ii) With reference to the role of the SAN, suggest how a slow rate of ventricular contraction could indicate that the SAN is **not** functioning correctly.

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(b) In people with Wolff–Parkinson–White syndrome, there is a bundle of fibres known as the bundle of Kent. These fibres can conduct impulses from the atria to the ventricles. This means that impulses do **not** always take the normal route through the AVN to the ventricles.

With reference to the role of the AVN, suggest **and** explain the change that occurs in the heart rate when impulses pass down the bundle of Kent to the ventricles instead of passing through the AVN.

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